1.08 Graph Sketching: Quadratic Graphs

Chapter 2 · Original slide 26 · 4: Quadratic Graphs

Recall that x refers to the input of a function, and the expression $f(x)$ refers to the output. For graph sketches, we often write $y = f(x)$, i.e. we set the y values to be the output of the function.

y

A 'sketch' in maths has a clearly defined meaning:

• We care about the general shape of the graph, not exact points.

• The axes should have no scale.

• Generally the only coordinates indicated are intercepts with the axis or other points of interest (e.g. intersections of multiple graphs)

Features needed in sketch of quadratic?

Recall a root of a function is where the output, in this case the y value, is 0.

Roots

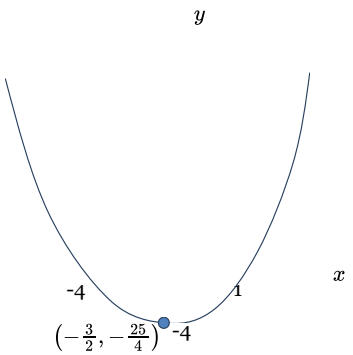
x

General shape: Smiley face or hill?

y-intercept

Turning point

Chapter 2 · Original slide 27 · Example



Sketch the graph of $y = x^2 + 3x - 4$ and find the coordinates of the turning point.

Roots:

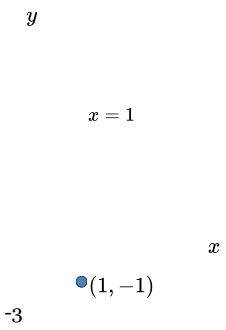
y-intercept:

Turning point:

Recall that if $f(x) = (x + a)^2 + b$, the minimum output is b and $-a$ is the x value which minimises it. i.e. Turning point is $(-a, b)$

1.08 Graph Sketching: Quadratic Graphs

Chapter 2 · Original slide 28 · Example



Sketch the graph of $y = 4x - 2x^2 - 3$ and find the coordinates of the turning point. Write down the equation of the line of symmetry.

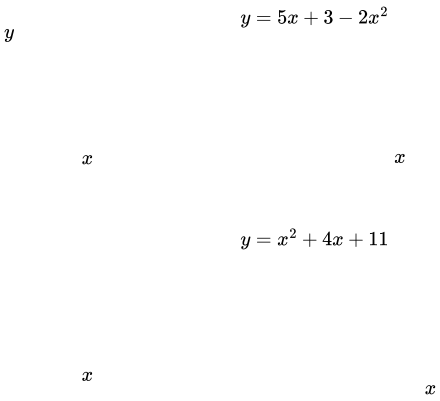
Roots:

y-intercept:

Turning point:

Line of reflection:

Chapter 2 · Original slide 29 · Test Your Understanding



Sketch the following, indicating any intercepts with the axis, the turning point and the equation of the line of symmetry.

a c

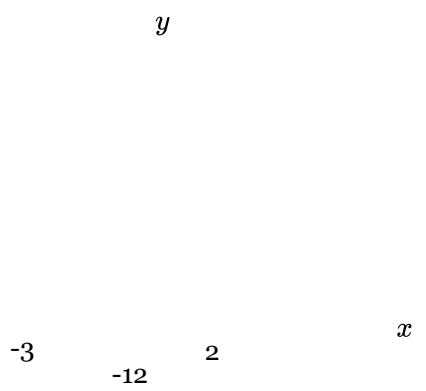
$y = x^2 + 4$

b d

$y = x^2 - 7x + 10$

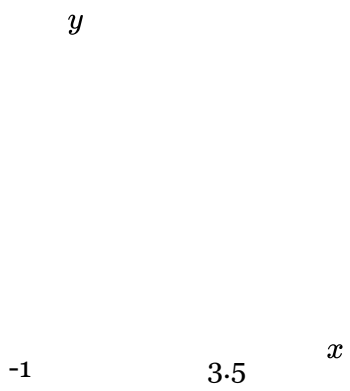
1.08 Graph Sketching: Quadratic Graphs

Chapter 2 · Original slide 30 · Determining the Equation using a Graph



Determine the equation of this quadratic graph, in the form $y = ax^2 + bx + c$.

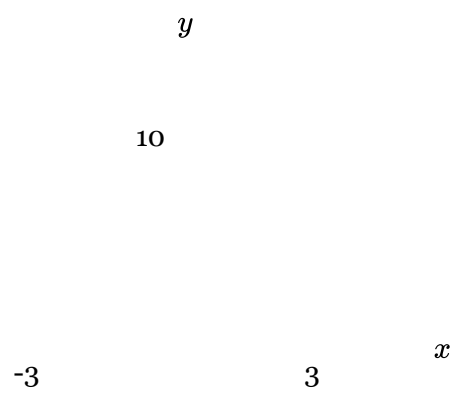
Chapter 2 · Original slide 31 · Further Example



Determine the equation of this quadratic graph, in the form $y = ax^2 + bx + c$, where a, b, c are integers.

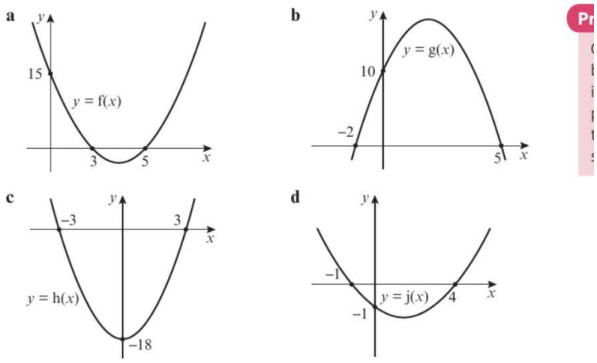
1.08 Graph Sketching: Quadratic Graphs

Chapter 2 · Original slide 32 · Test Your Understanding



Determine an equation of this quadratic graph.

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2 These sketches are graphs of quadratic functions of the form $ax^2 + bx + c$. Find the values of a,b,c for each function.

1.08 Graph Sketching: Quadratic Graphs

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3 The graph of $y = ax^2 + bx + c$ has a minimum at (5,-3) and passes through (4,0). Find the values of a,b,c.

[MAT 2003 1H] Into how many regions is the plane divided when the following three parabolas are drawn?

$y = x^2$
$y = x^2 - 2x$
$y = x^2 + 2x + 2$

x

Extension Question:

y